

# 1 HI-Jass User Manual

## 1.1 Running the application

### 1.1.1 Linux

`./run_hi_jass.sh`

### 1.1.2 Windows

Double-click `run_hi_jass.bat`, or run:

`.\run_hi_jass.bat`

The launcher creates the virtual environment, installs the dependencies, and starts the GUI.

## 1.2 Default NBI configuration

The default configuration contains two neutral beams:

| Beam  | Species | Injection energy | Injected power | Tangent $R_t$ | Direction |
|-------|---------|------------------|----------------|---------------|-----------|
| NBI-1 | D       | 120 keV          | 10 MW          | 0.5 m         | co        |
| NBI-2 | T       | 180 keV          | 0.1 MW         | 0.6 m         | counter   |

## 1.3 Two-beam slowing-down model

The slowing-down quantities are calculated independently for each beam. For beam  $j$ :

$$E_{c,j} = \left( \frac{m_j}{m_e} \right)^{1/3} T_e$$

$$\tau_{se,j} = 6.27 \times 10^8 \frac{A_j T_e [\text{eV}]^{3/2}}{n_e [\text{cm}^{-3}] \ln \Lambda}$$

$$\tau_{S,j} = \frac{\tau_{se,j}}{3} \ln \left[ 1 + \left( \frac{E_{b,j}}{E_{c,j}} \right)^{3/2} \right]$$

The useful beam power includes the HotJass capture and loss cascade. The fast-ion density from each beam is:

$$n_{b0,j} = \frac{P_{\text{useful},j} \tau_{S,j}}{E_{b,j} (10^3 e) V}$$

The total fast-ion density is:

$$n_{b0} = \sum_j n_{b0,j}$$

## 1.4 Density peaking and shine-through

The density peaking input defines the analytic electron-density profile:

$$n_e(\rho) = n_{e0}(1 - \rho^2)^{2p_n}, \quad 0 \leq \rho \leq 1$$

Shine-through uses the density along the beam chord rather than assuming that the central density applies everywhere. Each NBI supplies a tangent point  $(R_t, Z_t)$  in the NBI tab. The straight tangential chord at that point is intersected with the shaped plasma cross-section, and the optical depth is evaluated as:

$$\tau_b = \sigma_{\text{stop}}(E_b/A_b) \int_{\text{chord}} n_e(\rho(s)) ds$$

and the captured and shine-through fractions are:

$$f_{\text{capt},b} = 1 - e^{-\tau_b}, \quad f_{\text{shine},b} = e^{-\tau_b}.$$

The chord is evaluated numerically using the local cylindrical radius along the tangent path and the plasma elongation. With  $p_n = 0$ , the density is uniform along the computed chord; with  $p_n > 0$ , the profile is integrated point by point. A tangent point outside the plasma produces zero captured power.

### 1.4.1 Shine-through model selection

Each NBI has a three-way shine-through selector:

- **Riviere:** the existing Riviere-type analytic stopping cross-section fit.
- **Janev/Suzuki:** the Janev-Boley-Post analytic fit from the supplied paper, Nuclear Fusion 29 (1989) 2125, Eqs. (23)-(26), using the published Table III coefficients and the carbon-impurity coefficients from Table IV. It models total effective stopping, including the paper's density, electron-temperature and  $Z_{\text{eff}}$  dependence. The fit is published for  $100 \leq E \leq 10^4$  keV/u,  $10^{12} \leq n_e \leq 10^{15}$  cm<sup>-3</sup> and  $1 \leq T_e \leq 50$  keV; inputs outside these ranges are clipped to the nearest validity boundary.
- **Manual:** the entered manual shine-through fraction is used directly; captured power is  $(1 - f_{\text{shine}})P_{NB}$ .

The selected calculated stopping model is also used by the optional first-orbit-loss attenuation weighting. The Manual mode bypasses the cross-section calculation for shine-through.

## 1.5 First-orbit loss (reduced model)

First-orbit loss is an **opt-in** channel (the *First-orbit loss* checkbox in the Losses section). When enabled, it estimates the fraction of *captured* fast ions that leave the plasma promptly on their first drift orbit, before any slowing-down or charge-exchange because they are born close enough to the last closed flux surface (LCFS) that their guiding-centre orbit, or their gyro-orbit, crosses it.

The **Orbit model** selector (Losses section) chooses how the orbit widths and the loss criterion are built:

- **Large-aspect** ( $q_* \rho_{Li}$ ) the original reduced model described in *Passing-orbit width* and *Loss criterion* below. Direction-only mean shift; exactly zero co-current loss.
- **ST orbits mean-shift (arbitrary A)** and **ST orbits pitch-resolved** two spherical-tokamak variants described in *ST orbit models* at the end of this section. Both give non-zero co-current loss.

### 1.5.1 Birth profile along the chord

The calculation reuses the shine-through deposition. Along the straight tangential chord (assumed  $R_t = R_0$ , length  $2a$ ) the local ionisation rate density is

$$\dot{n}(x) \propto n_{e0} \sigma_{\text{stop}}(E_{b,j}/A_j) e^{-n_{e0} \sigma_{\text{stop}} x}, \quad x \in [0, 2a],$$

and the path coordinate is mapped to a normalised flux label by

$$\rho(x) = \frac{|x - a|}{a},$$

so  $\rho = 0$  at the chord mid-point and  $\rho = 1$  at both ends. The stopping cross-section is the beam's selected shine-through model; **Manual** mode has no deposition shape, so the Riviere fit is substituted for this weighting only.

**Off-axis / vertically-shifted injection.** When a beam's tangency radius  $R_t$  differs from  $R_0$ , or its aim height  $Z_t \neq 0$ , the birth profile is built instead from the *real* chord: a horizontal line tangent to the cylinder  $R = R_t$  at height  $Z_t$ , sampled through elliptical flux surfaces

$$\rho(s)^2 = \left( \frac{R(s) - R_0}{a} \right)^2 + \left( \frac{Z_t}{\kappa a} \right)^2, \quad R(s) = \sqrt{R_t^2 + (s - s_0)^2},$$

with the chord length running edge-to-edge and  $\rho$  no longer reaching 0 (the beam no longer passes through the magnetic axis). This is the same geometry the shine-through / capture calculation already uses. For the standard on-axis tangential case ( $Z_t = 0$ ,  $R_t = R_0$  or unset) the crude  $\rho(x) = |x - a|/a$  mapping above is kept, so existing results are unchanged.

**Central column.** The beam enters from the outboard side. If its tangency radius is inside the central column ( $R_t < R_{\text{post}}$ , set by **Centre-post** R; default  $R_0 - a$ ), the straight chord would strike the column, so only the **outboard leg** from the outboard-edge entry to the point where  $R(s) = R_{\text{post}}$  is kept; the (unreachable) far leg is discarded. A beam with  $R_t$  at or outboard of  $R_{\text{post}}$  threads the gap between the column and the inboard plasma edge and keeps its full chord. A beam whose aim point lies outside the plasma altogether contributes no first-orbit loss. The Deposition tab's geometry panel draws the column and marks a blocked beam.

### 1.5.2 Passing-orbit width

The radial width of a strongly co-passing drift orbit at the full injection energy is the textbook large-aspect-ratio estimate

$$\Delta r_j = q_* \rho_{Li,j}, \quad q_* = \frac{2\pi a^2 B_t}{\mu_0 R_0 I_p} \frac{1 + \kappa^2}{2}, \quad \rho_{Li,j} = \frac{m_j v_{b,j}}{e B_t}.$$

$q_*$  is the cylindrical, elongation-corrected safety factor and  $\rho_{Li,j}$  the toroidal-field gyroradius at  $E_{b,j}$ .

### 1.5.3 Loss criterion

The orbit is modelled as a **rigid mean shift** of the birth radius: inward for co-current injection (favourable), outward for counter-current:

$$\rho_{\text{eff}}(x) = \rho(x) \mp \frac{\Delta r_j}{a} \quad (\text{co / counter}),$$

and an ion born at  $x$  is counted as lost if  $\rho_{\text{eff}}(x) > 1$ . Because  $\rho(x) \leq 1$ , the co-current shift can never trigger this test, so the mean-shift model gives **exactly zero first-orbit loss for co-current injection**; for counter-current it loses every ion born with  $\rho(x) > 1 - \Delta r_j/a$ .

An optional, direction-independent **Larmor (gyro-orbit) prompt-loss** channel can be OR-ed in: an ion whose guiding centre is born within one gyro-diameter of the LCFS,

$$\rho(x) > 1 - \frac{2\rho_{Li,j}}{a},$$

is lost regardless of the drift direction (gyration is much faster than the guiding-centre drift). This channel is disabled by default.

### 1.5.4 Loss fraction and power

$$f_{\text{orbit},j} = \frac{\int_{\text{lost}}^x \dot{n}(x) dx}{\int_0^{2a} \dot{n}(x) dx}, \quad P_{\text{orbit},j} = f_{\text{orbit},j} P_{\text{capt},j}.$$

First-orbit loss is applied to the captured power *before* the charge-exchange loss and before the useful power that drives  $T_e$ ,  $T_i$  and  $n_{b0,j}$ . Injection direction (co-/counter-current) is set per NBI in its input section.

### 1.5.5 Reported quantities

The Dashboard and the Assumptions tab report, per beam,  $\rho_{Li,j}$  and  $\Delta r_j$  (in cm and as fractions of  $a$ ) and  $f_{\text{orbit},j}$ .

### 1.5.6 Limitations

- **The co-current zero is a model artefact, not physics.** The rigid mean-shift ignores the finite radial width of the orbit and the outboard excursion that a favourable inward mean shift does not cancel; published orbit-following comparisons give of order a few percent co-current first-orbit loss, versus tens of percent counter-current.
- The chord label  $\rho(x) = |x - a|/a$  folds the inboard and outboard halves together, so the model cannot distinguish a high-field-side birth from a low-field-side birth. It therefore cannot represent the correct structure co-current losing high-field-side edge births, counter-current losing low-field-side edge births, and trapped-ion banana-tip losses on the low-field side in either direction.
- Circular / elliptical cross-section, large-aspect-ratio orbit widths (the birth chord itself now honours  $R_t$  and  $Z_t$ , but the orbit-width and pitch estimates are still evaluated at  $R_0$ );  $\Delta r_j$  and  $\rho_{Li,j}$  are evaluated at the full injection energy (an upper bound real ions slow down).
- The birth-profile integral in this channel uses the flat central density  $n_{e0}$ , not the peaked profile used elsewhere.

- When  $\rho_{Li}/a$  or  $\Delta r/a \gtrsim 0.3$  orbit width comparable to the machine size, typical of compact low-aspect-ratio devices the whole estimate is order-unity uncertain and really needs a guiding-centre orbit follower. The Assumptions tab flags this case. The ST orbit models below address the first three of these limitations directly.

### 1.5.7 ST orbit models (arbitrary aspect ratio)

On a spherical tokamak the drift and banana orbit widths are an  $O(a)$  fraction of the minor radius, the trapped fraction is large, and outboard-born trapped ions are lost **regardless of injection direction** so the large-aspect model's exact co-current zero is qualitatively wrong. The two ST variants build their widths from the **poloidal gyroradius**

$$\rho_{\theta,j} = \frac{B_t}{\bar{B}_p} \rho_{Li,j}, \quad \bar{B}_p = \frac{\mu_0 I_p}{2\pi a \sqrt{(1 + \kappa^2)/2}},$$

(not  $q_* \rho_{Li}$ , whose  $q_*$  runs away as  $\varepsilon \rightarrow 1$ ), together with the circulating and banana widths and the arbitrary- $A$  trapped fraction

$$w_{\text{pass},j} = \varepsilon \rho_{\theta,j}, \quad w_{\text{ban},j} = 2\sqrt{\varepsilon} \rho_{\theta,j}, \quad f_t = 1 - \frac{(1 - \varepsilon)^2}{(1 + 1.46\sqrt{\varepsilon})\sqrt{1 - \varepsilon^2}},$$

with  $\varepsilon = a/R_0$  (clamped to  $[10^{-3}, 0.95]$ ) and  $f_c = 1 - f_t$  (Lin-Liu & Miller, *Phys. Plasmas* **2** (1995) 1666). An arbitrary- $A$  edge safety factor  $q_a$  (Uckan / ITER Physics Basis form, with the  $(1.17 - 0.65\varepsilon)/(1 - \varepsilon^2)^2$  shaping factor) is reported for reference but is **not** used in the widths.

The birth profile in this channel uses the same Beer-Lambert deposition density as the large-aspect model and the Deposition tab's chord plot,  $n_e \sigma e^{-n_e \sigma x}$  via the selected stopping model so it responds to  $n_{e0}$  the same way everywhere in the code (an earlier version used a fixed "penetration rule of thumb" mean free path that saturated at its clamp for realistic densities, decoupling the ST models' loss fraction from  $n_{e0}$ ; this has been fixed).

The **drift-orbit shift is direction-dependent** for both populations inward for co-current ( $v_{\parallel}$  along  $I_p$ ), outward for counter with a weaker shift  $c_t w_{\text{pass}}$  ( $c_t = 0.25$ ) for the trapped-ion banana centre than for a passing ion. The **gyro channel** ( $\rho > 1 - \rho_{Li}/a$ , born within one gyroradius of the LCFS) is direction-independent. Co-current loss is therefore non-zero but smaller than counter-current the co/counter asymmetry survives, unlike the large-aspect model's all-or-nothing 0 vs 1.

**Mean-shift variant.** Deposition-weighted loss integrals over  $\rho(x)$ , blended by  $f_c, f_t$  (sign convention:  $-$  co,  $+$  counter):

$$f_{\text{orbit},j} = f_c L_{\text{pass}} + f_t L_{\text{trap}},$$

$$L_{\text{pass}} : \rho \mp \frac{w_{\text{pass},j}}{a} > 1 \quad \text{or} \quad \rho > 1 - \frac{\rho_{Li,j}}{a},$$

$$L_{\text{trap}} : \rho + \frac{w_{\text{ban},j}}{2a} \mp \frac{0.25 w_{\text{pass},j}}{a} > 1 \quad \text{or} \quad \rho > 1 - \frac{\rho_{Li,j}}{a}.$$

**Pitch-resolved variant.** A 2-D integral over birth radius  $x$  and the pitch **magnitude**  $|\lambda| = |v_{\parallel}/v|$ . The birth value  $|\lambda_0(x)| = R_t/\sqrt{R_t^2 + (x - a)^2}$  is fixed by the injection *geometry* and does **not** depend on the poloidal-field / current direction only the sign of  $v_{\parallel}$  relative to  $I_p$  flips (co

$\leftrightarrow$  counter), carried by  $s_{\text{dir}} = \mp 1$ .  $|\lambda_0|$  is smeared by a  $\sigma = 0.10$  Gaussian. An ion is trapped if  $|\lambda| < \sqrt{2\varepsilon_{\text{loc}}/(1 + \varepsilon_{\text{loc}})}$  ( $\varepsilon_{\text{loc}} = \rho a/R_0$ ); its outboard radial excursion is

$$\text{passing : } s_{\text{dir}} \frac{w_{\text{pass},j}}{a} |\lambda|, \quad \text{trapped : } \frac{w_{\text{ban},j}}{2a} \frac{|\lambda|}{\lambda_{\text{tr}}} + s_{\text{dir}} \frac{0.25 w_{\text{pass},j}}{a},$$

(the banana half-width  $\rightarrow 0$  for deeply trapped ions, largest near the trapped/passing boundary), and it is lost if  $\rho + (\text{excursion}) > 1$  or  $\rho > 1 - \rho_{Li,j}/a$ .

Both variants collapse toward the large-aspect co-current zero as  $\varepsilon \rightarrow 0$ . They remain 0-D order-of-magnitude estimates absolute magnitudes are upper-bound-like and, because the direction-independent trapped banana-tip and gyro channels stay significant, the co/counter split is narrower than the large-aspect model's (typically a factor  $\sim 1.5$ , not 0 vs 1). References: Akers *et al.*, *Nucl. Fusion* (START NBI,  $A \sim 1.4$ ); Goldston, White & Boozer, *Phys. Rev. Lett.* **47** (1981) 1004; Goldston & Rutherford, *Introduction to Plasma Physics* (1995), Ch. 12.

## 1.6 Charge-exchange loss

A captured fast ion can charge-exchange with a background cold neutral during slowing-down, becoming a fast **neutral** that escapes the plasma before fully thermalizing a second loss channel, applied to the post-orbit-loss captured power (Step 0, same ordering as first-orbit loss above). Design note: docs/CX model.pdf. A **CX-loss model** selector (rail, "Losses" section) picks between three modes; only the active mode's inputs matter.

### 1.6.1 Manual fraction (default)

The original flat efficiency knob `cx_loss_fraction`  $\in [0, 1]$ : a fixed fraction of captured power is removed, independent of  $E_b$ ,  $n_e$ , or any background-neutral density. PPPL-1280 Sect. 6.2.2/6.2.3 quotes "up to 10%" of injected power for  $Z_{\text{eff}} = 1$  as a rough qualitative figure, not a formula. Default 0=off; leaves every other result byte-identical.

### 1.6.2 Manual $n_0/n_e$ the physics-based survival integral

The user supplies  $n_0/n_e$  directly (a spatially uniform, 0-D background neutral density  $n_0 = (n_0/n_e)n_{e0}$ ), and the model evaluates the actual CX survival integral along the slowing-down path. For a fast ion at energy  $E$ :

$$\nu_{\text{cx}}(E) = n_0 \sigma_{\text{cx}}(E) v(E), \quad S(E) = \exp \left[ - \int_E^{E_b} \frac{\nu_{\text{cx}}(E')}{|dE'/dt|} dE' \right],$$

with particle- and power-loss fractions

$$f_{\text{cx},N} = 1 - S(0^+), \quad f_{\text{cx},P} = \int_0^{E_b} \frac{E}{E_b} \frac{\nu_{\text{cx}}(E)}{|dE/dt|} S(E) dE.$$

$f_{\text{cx},P}$  (the quantity actually applied to  $P_{\text{useful}}$ ) is smaller than  $f_{\text{cx},N}$  because  $\sigma_{\text{cx}}$  falls steeply above  $\sim 3050$  keV/amu, so most CX events happen late in the cascade, when little energy is left to remove.  $1/|dE/dt| = (\tau_{se}/2)\sqrt{E}/(E^{3/2} + E_c^{3/2})$  is exactly the same slowing-down kernel already used by `thermalization_time/slowing_down_distribution` (differentiating  $\tau_s(E) = (\tau_{se}/3) \ln[1 + (E/E_c)^{3/2}]$  reproduces it), so no new slowing-down physics is introduced only the extra  $\nu_{\text{cx}}(E)$  factor under the same integral.  $\sigma_{\text{cx}}(E)$  is the Janev & Smith (1993, Sect. 2.3.1)

analytic form, refit by Swaczyna, Bzowski & Kubiak (arXiv:2411.13174, 2024, their Eq. A1) against Schultz *et al.* (2023) theory and the Barnett (1990) tabulation quoted accuracy 5% from 1100 keV/amu, the range this project’s beams live in.

Also reported, per beam, on the Dashboard: the dimensionless figure of merit  $\gamma_{\text{cx}} \equiv \nu_{\text{cx}}(E_b) \tau_s \approx n_0 \sigma_{\text{cx}}(E_b) v_b \tau_s$  negligible for  $\gamma_{\text{cx}} \ll 1$ , worth taking seriously for  $\gamma_{\text{cx}} \sim 0.11$ , and a sign the single-pass loss treatment is breaking down for  $\gamma_{\text{cx}} \gtrsim 1$  (flagged with a warning, as is  $f_{\text{cx},P} > 10\%$ , PPPL-1280’s rough  $Z_{\text{eff}} = 1$  ceiling). Because CX loss depends on  $T_e$  (via  $E_c(T_e)$ ) while  $T_e$  itself depends on the CX-reduced  $P_{\text{useful}}$ , this is a genuine fixed point; an outer under-relaxed Picard loop (in `hotjass_core`, mirroring the existing alpha-heating fixed point) closes it, keeping the core solver a single non-iterative solve.

### 1.6.3 Penetration (n0\_LCFS/ne) edge-penetration estimate

Instead of a single uniform  $n_0$ , the boundary value  $n_{0,\text{LCFS}} = (n_{0,\text{LCFS}}/n_e) n_{e,\text{LCFS}}$  is set from the user ratio, and the core profile follows the standard penetration/recycling picture.  $n_{e,\text{LCFS}}$  is evaluated at  $\rho = 0.95$ , not literally at  $\rho = 1$ : the core density shape  $n_e(\rho) = n_{e0}(1 - \rho^2)^{2p_n}$  is an exact zero at  $\rho = 1$  whenever  $p_n > 0$ , which would make  $n_{0,\text{LCFS}} = 0$  regardless of the input ratio – a knob that does nothing.  $\rho = 0.95$  is a compromise: with a strongly peaked profile ( $p_n \gtrsim 12$ ) it can still sit orders of magnitude below  $n_{e0}$ , so treat this mode as unreliable for very peaked density profiles.

$$\lambda_0(\rho) = \frac{v_0}{n_e(\rho) [\langle \sigma v \rangle_{\text{ion}}(T_e) + \langle \sigma v \rangle_{\text{cx}}(T_i)]}, \quad n_0(\rho) \approx n_{0,\text{LCFS}} \exp \left[ - \int_{\rho}^1 \frac{a}{\lambda_0(\rho')} d\rho' \right],$$

volume-averaged to a single  $\langle n_0 \rangle$  and fed through the same survival integral as ”Manual n0/ne” above.  $v_0$  is a round  $\sim 3$  eV Franck-Condon-like dissociation speed. **Caveat:** the ionization ( $\langle \sigma v \rangle_{\text{ion}}(T_e)$ ) and thermal-CX ( $\langle \sigma v \rangle_{\text{cx}}(T_i)$ ) rate coefficients used for  $\lambda_0$  are an interim hand-built approximation (a threshold-shaped form for ionization, a threshold-free  $\sqrt{T}$  form for CX, each capped to saturate at a few  $\times 10^{-14}$  m<sup>3</sup>/s) a calibrated Voronov (1997) / ADAS `adf11` fit was not sourced/verified in this session, so treat  $n_0(\rho)$ ’s absolute scale as order-of-magnitude only; the survival-integral physics downstream of  $\langle n_0 \rangle$  (same  $\sigma_{\text{cx}}(E)$  fit as ”Manual n0/ne”) is not affected by this caveat.  $n_0(\rho)$  itself (per the active mode – flat for ”manual n0/ne”, the profile above for ”penetration”, nothing for ”manual fraction”) is plotted on the Profiles tab; because  $\lambda_0$  is typically much smaller than  $a$ , the profile can span many orders of magnitude edge-to-core (shown on a log axis) – a genuine, if extreme, feature of this reduced model, not a numerical artifact.

### 1.6.4 Escape probability (informational only)

A fast neutral born from a CX event can re-ionize before reaching the wall, so treating every CX event as a full loss is a conservative upper bound, not exact. An escape-probability factor  $\exp(-x_{\text{wall}}/\lambda_{\text{reion}})$  is shown per beam (Dashboard)  $x_{\text{wall}} \approx a$ ,  $\lambda_{\text{reion}}$  from the same combined ionization+CX stopping cross section a beam neutral itself uses but is **not** applied to  $f_{\text{cx},P}$ ; it is informational, per the design note’s explicit ”acceptable as the conservative default, but shown as such.”

## 1.7 Bulk toroidal rotation

A **Rotation model** selector (Losses section, below the CX-loss fields) picks between three modes, off by default – like every other loss/model knob, this leaves every existing result byte-identical

unless explicitly enabled.

**Where rotation matters and where it doesn't.**  $v_\phi$  (typically  $\sim 10^2 10^3$  km/s) is small next to the beam's own velocity ( $\sim 35 \times 10^3$  km/s for a 100180 keV D/T beam), so shine-through, first-orbit loss and CX loss – all governed by the beam's own velocity – are essentially insensitive to it and are **not** touched. The one channel where it matters cleanly is **beam-target fusion**: that reaction's cross-section and flux factor depend on the beam-target *relative* velocity, which the un-rotated model implicitly assumes equals the beam's lab velocity (a stationary target). With rotation:

$$v_{\text{rel}}(E) = |v_{\text{beam}}(E) - s v_\phi|, \quad s = +1 \text{ (co-current beam), } -1 \text{ (counter),}$$

used for *both* the flux factor and the cross-section's energy argument (re-expressed as the equivalent deuteron energy  $E_D = \frac{1}{2} M_D v_{\text{rel}}^2$ , the same conversion `beam_target_reactivity_spectrum()` already used for a T beam).  $v_\phi = 0$  reproduces the original stationary-target result exactly.

**Non-monotonic, not just a small correction in one direction.** The D-T cross-section peaks around  $\sim 64$  keV (deuteron-equivalent energy) and falls off on both sides. For a beam energy *above* that peak (this project's 120180 keV presets all are), pulling  $v_{\text{rel}}$  down (co-current beam, co-rotating plasma) moves the *high-energy tail* of the slowing-down spectrum back toward the peak and can *increase* beam-target yield, while a counter-current beam sees the opposite – the sign is not "co always suppresses fusion", it depends on where  $E_b$  sits relative to the cross-section peak, and the net effect is an integral over the whole slowing-down spectrum (not just  $E_b$  itself), evaluated numerically.

### 1.7.1 Manual $v_\phi$

Direct input of a single, spatially-uniform  $v_\phi$  (m/s, positive = co-current) – the honest option when the collisionality of the source is unknown but a plausible rotation speed is.

### 1.7.2 Momentum balance

$v_\phi$  instead comes from a 0-D toroidal angular-momentum balance, the rotation analogue of this project's energy balance:

$$T_{\text{NBI}} = \sum_j s_j \frac{2P_{\text{useful},j} R_{t,j}}{v_{b,j}}, \quad v_\phi = \frac{T_{\text{NBI}} \tau_\phi}{\rho_i R_0 V},$$

summing each beam's injected canonical angular momentum ( $s_j = +1$  co,  $-1$  counter;  $R_{t,j}$  its own tangency radius), balanced against a momentum-confinement loss  $L/\tau_\phi$  with  $L \sim \rho_i v_\phi R_0 V$  ( $\rho_i$  = thermal-ion mass density,  $n_{D0} M_D + n_{T0} M_T$ ) – the bulk plasma treated as a single representative rotating mass at the magnetic axis, the same level of 0-D reduction as everywhere else in this model. **Caveat:** unlike  $\tau_E$ , there is no widely validated scaling for the momentum confinement time  $\tau_\phi$ ; the user-set ratio `tau_phi / tauE,i` (default 1.0, the common crude  $\tau_\phi \approx \tau_{E,i}$  assumption) sets  $\tau_\phi = (\text{ratio}) \times \tau_{E,i}$ , so treat  $v_\phi$ 's absolute scale here as order-of-magnitude, more so than the manual-input mode.

The Dashboard reports  $v_\phi$  (km/s) and a Mach number  $M = v_\phi/c_s$  ( $c_s = \sqrt{(T_{e0} + T_{i0})/m_i}$ , a simple ion-acoustic estimate, not a rigorous sound speed), plus the net torque  $T_{\text{NBI}}$  for the momentum-balance mode.

## 1.8 Beam-beam fusion

Off by default (checkbox, Losses section, below the rotation fields). Fast ions from **two separate NBI sources** can fuse with each other directly – a third fusion channel alongside thermal-thermal and beam-target. This is a **reduced** model, not a full treatment: instead of integrating over both beams’ slowing-down spectra (a genuine 2-D relative-velocity integral), each population is approximated by a single characteristic velocity at its own average fast-ion energy (`average_fast_energy_keV`, the same representative energy this project already uses for pressure/energy-density diagnostics), signed by its own co/counter direction. Only the **first two** useful beams are paired – this project’s tested scope is exactly two NBI sources.

The relevant physical quantity is the beam-beam **relative** velocity:

$$v_{\text{rel}} = |s_1 v_1 - s_2 v_2|, \quad s = +1 \text{ (co)}, -1 \text{ (counter)},$$

fed into the same deuteron-equivalent-energy cross-section evaluation as the rotation correction to beam-target fusion (Sec. “Bulk toroidal rotation” above) – structurally the same construction, with the *other beam’s own velocity* playing the role  $v_\phi$  played there. **This is why counter-injecting the second beam matters physically:** two co-current beams at similar per-amu energy have a small  $v_{\text{rel}}$  (their velocities nearly cancel) and thus a negligible beam-beam yield, while opposite directions maximize  $v_{\text{rel}}$  – exactly DANTE’s default configuration (D co-current, T counter-current). D-T and D-D pairs are handled (the only cross-section fits this project carries); T-T pairs, or a beam-beam calculation with fewer than two useful beams, contribute nothing.

The result is folded into the existing D-T/D-D fusion power and neutron- rate totals (`pf_dt_w`, `pf_dd_w`, `neutron_rate_s`) exactly like the thermal and beam-target contributions, so it is automatically included in the alpha-heating closure (Sec. “Alpha self-heating” below) with no separate wiring needed. The Dashboard’s D-T/D-D fusion-power rows show all three contributions (thermal / beam-target / beam-beam); the Scan tab’s “Neutrons” group breaks the neutron rate out the same way, with the beam-beam panel showing a genuine (if usually small, for DANTE’s default 0.1 MW counter-beam) nonzero curve once enabled, rather than the flat zero it shows while off.

## 1.9 Power balance (electron and ion)

At each central density the model solves a 0-D steady-state power balance for  $T_e$  and  $T_i$ . The heating that drives each temperature is the sum of the neutral-beam split, prescribed auxiliary heating (ECRH and ICRH), and when enabled fusion alpha self-heating.

### 1.9.1 Beam heating split

Each useful beam  $j$  delivers a fraction  $L_e(x_j)$  of its power to electrons and  $L_i(x_j) = 1 - L_e(x_j)$  to ions, with  $x_j = E_{b,j}/E_{c,j}(T_e)$  and

$$L_i(x) = \frac{1}{x} \int_0^x \frac{du}{1 + u^{3/2}}, \quad L_e(x) = 1 - L_i(x).$$

$L_i \rightarrow 1$  as  $x \rightarrow 0$  (cold, ion-dominated) and  $L_i \rightarrow 0$  as  $x \rightarrow \infty$  (fast beam, electron-dominated).  $P_{\text{useful},j}$  is the injected power after shine-through, first-orbit loss and charge-exchange loss (preceding sections).

### 1.9.2 Auxiliary heating (ECRH, ICRH)

ECRH and ICRH are entered as powers with electron/ion fractions. Their contribution to each channel is

$$P_{\text{aux},e} = f_e^{\text{EC}} P_{\text{ECRH}} + f_e^{\text{IC}} P_{\text{ICRH}}, \quad P_{\text{aux},i} = (1 - f_e^{\text{EC}}) P_{\text{ECRH}} + f_i^{\text{IC}} P_{\text{ICRH}}.$$

ECRH is electron-dominant,  $f_e^{\text{EC}}$  close to 1. For ICRH the electron and ion fractions are set independently; if  $f_e^{\text{IC}} + f_i^{\text{IC}} < 1$  the remainder is treated as power that does not thermalise (a fast-ion tail or direct loss). These sources are Te/Ti-independent constants added to the right-hand side of each balance.

### 1.9.3 Alpha self-heating (optional)

When enabled, 3.5 MeV fusion alphas deposit

$$P_\alpha = f_\alpha \frac{E_\alpha}{E_f} P_{f,\text{tot}} = f_\alpha \frac{3.5}{17.6} P_{f,\text{tot}},$$

split between electrons and ions by the alpha slowing-down fraction

$$L_e^\alpha = L_e \left( \frac{E_\alpha}{E_c^\alpha(T_e)} \right), \quad E_c^\alpha = \left( \frac{m_\alpha}{m_e} \right)^{1/3} T_e,$$

so  $P_{\alpha,e} = L_e^\alpha P_\alpha$  and  $P_{\alpha,i} = (1 - L_e^\alpha) P_\alpha$ . Because  $P_\alpha$  depends on  $P_{f,\text{tot}}$ , which depends on  $T_i$ , which depends on  $P_{\alpha,i}$ , the alpha term is closed by an under-relaxed fixed-point iteration; with alpha heating off the solve is a single pass.

### 1.9.4 Decoupled balance (default)

With electron-ion equipartition off, the two balances are solved in sequence. The electron balance is a 1-D root-find for  $T_e$ :

$$\frac{3}{2} n_{e0} T_e (10^3 e) \frac{V}{\tau_{E,e}} = \sum_j L_e(x_j) P_{\text{useful},j} + P_{\text{aux},e} + P_{\alpha,e}.$$

The ion temperature then follows in closed form (or, for the neoclassical-ion option, from a second 1-D root-find see *Confinement time*):

$$\frac{3}{2} n_{\text{thermal}} T_i (10^3 e) \frac{V}{\tau_{E,i}} = \sum_j L_i(x_j) P_{\text{useful},j} + P_{\text{aux},i} + P_{\alpha,i},$$

where  $n_{\text{thermal}} = n_{\text{sum}}(n_{e0}, Z_{\text{eff}}) - n_{b0}$  is the thermal-ion density left after the fast ions are subtracted. A non-positive  $n_{\text{thermal}}$  marks the operating point infeasible.

### 1.9.5 Coupled balance (equipartition on)

When electron-ion equipartition is enabled, the exchange power  $P_{ei}(T_e, T_i)$  (defined in *Electron-ion exchange time*) is subtracted from the electron balance and added to the ion balance, and the pair is solved together:

$$\frac{3}{2}n_{e0}T_e(10^3e)\frac{V}{\tau_{E,e}} = \sum_j L_e(x_j)P_{\text{useful},j} + P_{\text{aux},e} + P_{\alpha,e} - P_{ei}(T_e, T_i),$$

$$\frac{3}{2}n_{\text{thermal}}T_i(10^3e)\frac{V}{\tau_{E,i}} = \sum_j L_i(x_j)P_{\text{useful},j} + P_{\text{aux},i} + P_{\alpha,i} + P_{ei}(T_e, T_i).$$

The reported  $P_e$  and  $P_i$  remain the pure beam split;  $P_{ei}$ ,  $P_{\alpha}$  and  $P_{\text{aux}}$  are reported separately.

### 1.9.6 Profile-corrected 0-D

The balance above is genuinely 0-D: one density, one  $T_e$ , one  $T_i$ , treated as uniform over the volume. With the **profile-corrected 0-D** checkbox (Plasma section) *off* the default the entered central density is used directly as that uniform value and the solved  $T$  is a flat-plasma effective temperature; comparing it to a measured on-axis  $T_0$  then needs a peaking factor supplied by the reader (validation runs use  $\simeq 2$ ).

With the checkbox *on*, the entered **central n.e** is taken to be the **on-axis**  $n_{e0}$ , and for a  $(1-\rho^2)^{2p}$  profile the volume average is  $\langle X \rangle = X_0/(1+2p)$ . The energy balance and the  $\tau_E$  scalings then run on

$$\langle n_e \rangle = \frac{n_{e0}}{1+2p_n},$$

so the solved  $T_e$ ,  $T_i$  are **volume-averaged**. The electron and ion temperature profiles are shaped by **independent** exponents  $p_{Te}$  (**Temp peaking (electron)**) and  $p_{Ti}$  (**Temp peaking (ion)**); a value  $< 0$  means "use  $p_{Te}$ "), so the on-axis values are

$$T_{e0} = \langle T_e \rangle (1+2p_{Te}), \quad T_{i0} = \langle T_i \rangle (1+2p_{Ti}).$$

They are reported (Dashboard row "on-axis  $T_{e0}/T_{i0}$ ") and are what the fusion, pressure and  $\beta_t$  integrals use as their central values the reactivity integrals with the  $(1-\rho^2)^{2p_{Ti}}$  ion shape, the beam slowing-down kernel with the  $(1-\rho^2)^{2p_{Te}}$  electron shape. Set  $p_n$  (**Density peaking**),  $p_{Te}$  and  $p_{Ti}$  to match the real profiles;  $p \simeq 0.5075$  gives  $X_0/\langle X \rangle \simeq 22.5$  (L-/H-mode), while a **hot-ion mode** on a compact ST needs  $p_{Ti} \gg p_{Te}$  ( $T_{i0}/\langle T_i \rangle \simeq 34$  with  $T_e$  nearly flat). With all peaking parameters at 0 the correction is a no-op.

### 1.10 Confinement time

$\tau_{E,e}$  and  $\tau_{E,i}$  in the balances above are set per channel by the *Confinement* selector in the Losses section; the electron and ion channels are chosen independently. When a physics-based scaling is selected the manual **tauE,e** / **tauE,i** inputs are ignored, and the Summary sheet reports the computed value (or its range across a scan) instead.

The physics-based scalings below all use the same loss power

$$P_{\text{loss}} = \sum_j P_{\text{useful},j} + P_{\text{aux},e} + P_{\text{aux},i},$$

i.e. the captured, post-orbit, post-charge-exchange beam power plus the prescribed auxiliary (ECRH, ICRH) heating. Fusion alpha heating is not included in  $P_{\text{loss}}$  (it is closed by an outer fixed-point and is not known when  $\tau_E$  is evaluated). None of these depend on  $T_e$  or  $T_i$ , so each is evaluated once.

### 1.10.1 Fixed (input)

$\tau_{E,e}$  and  $\tau_{E,i}$  are the values entered in the Losses section (under the Confinement selector), used unchanged.

### 1.10.2 IPB98(y,2) ELMy H-mode

The ITER Physics Basis ELMy H-mode scaling (ITER Physics Basis, *Nucl. Fusion* **39** (1999) 2175; standard reference form):

$$\tau_E = 0.0562 I_p^{0.93} B_t^{0.15} n_{19}^{0.41} P_{\text{loss}}^{-0.69} R_0^{1.97} \kappa_a^{0.78} \varepsilon^{0.58} M_{\text{eff}}^{0.19}$$

with  $I_p$  in MA,  $B_t$  in T,  $P_{\text{loss}}$  in MW,  $R_0$  and  $a$  in m,  $n_{19} = n_{e0}/10^{19} \text{ m}^{-3}$ ,  $\varepsilon = a/R_0$ ,  $\kappa_a = \kappa$ , and  $M_{\text{eff}} = 2x_D + 3x_T$  the mass-weighted ion mass number. Selecting *IPB98(y,2) ELMy H-mode* uses it for both channels ( $\tau_{E,e} = \tau_{E,i}$ ); *IPB98(y,2) e / neoclassical i* uses it for the electron channel only, with the ion channel from the neoclassical estimate below.

The fit was obtained from conventional-aspect-ratio devices ( $A \sim 2.5$ -4) and is documented in the spherical-tokamak literature to mis-predict at low aspect ratio. When  $A = R_0/a < 2$  the Assumptions tab flags that IPB98(y,2) is being extrapolated outside its dataset.

### 1.10.3 Kaye NSTX L-mode

An NSTX low-aspect-ratio (spherical tokamak) L-mode global fit (Kaye, *Nucl. Fusion* **46** (2006) 848, Eq. 5 / Fig. 9):

$$\tau_E = 4.73 \times 10^{-4} I_p^{1.01} B_t^{0.70} n_e^{-0.07} P_{\text{loss}}^{-0.37}$$

with  $I_p$  in A,  $B_t$  in T,  $n_e$  in  $\text{m}^{-3}$ ,  $P_{\text{loss}}$  in W and  $\tau_E$  in s. Selecting *Kaye NSTX L-mode* uses it for both channels ( $\tau_{E,e} = \tau_{E,i}$ ); *Kaye L e / neoclassical i* uses it for the electron channel only.

### 1.10.4 Kaye NSTX H-mode

The H-mode counterpart of the previous fit: an NSTX low-aspect-ratio (spherical tokamak) H-mode thermal-energy scaling (Kaye, *Nucl. Fusion* **46** (2006) 848, Table 1, ordinary-least-squares "Case 1" fit to all 85 H-mode points, RMSE = 0.145):

$$\tau_E = 4.69 \times 10^{-9} I_p^{0.57} B_t^{1.08} n_e^{0.44} P_{\text{loss}}^{-0.73}$$

with  $I_p$  in A,  $B_t$  in T,  $n_e$  in  $\text{m}^{-3}$ ,  $P_{\text{loss}}$  in W and  $\tau_E$  in s. Relative to IPB98(y,2) the current dependence is much weaker ( $I_p^{0.57}$  vs  $I_p^{0.93}$ ) and the toroidal-field dependence much stronger ( $B_t^{1.08}$  vs  $B_t^{0.15}$ ), the qualitative spherical-tokamak trend; the density and heating-power exponents ( $n_e^{0.44}$ ,  $P_{\text{loss}}^{-0.73}$ ) are of the same order as the conventional-aspect scalings. This is an ST-appropriate fit ( $A \sim 1.31.5$  dataset). Selecting *Kaye NSTX H-mode* uses it for both channels ( $\tau_{E,e} = \tau_{E,i}$ ); *Kaye H e / neoclassical i* uses it for the electron channel only.

### 1.10.5 Neoclassical ion

A banana-regime order-of-magnitude estimate (Wesson, *Tokamaks*; Chang-Hinton low-collisionality limit) for the ion channel only:

$$\chi_{i,\text{neo}} \simeq \frac{q^2 \rho_i^2 \nu_{ii}}{\varepsilon^{3/2}}, \quad \tau_{E,i} = \frac{a^2}{\chi_{i,\text{neo}}} = \frac{a^2 \varepsilon^{3/2}}{q^2 \rho_i^2 \nu_{ii}},$$

with  $\varepsilon = a/R_0$ , the elongation-corrected cylindrical safety factor

$$q = \frac{5 a^2 B_t}{R_0 I_p [\text{MA}]} \frac{1 + \kappa^2}{2},$$

the thermal-ion Larmor radius

$$\rho_i = \frac{m_{\text{eff}} v_{th,i}}{e B_t}, \quad v_{th,i} = \sqrt{\frac{2 T_i (10^3 e)}{m_{\text{eff}}}}, \quad m_{\text{eff}} = x_D m_D + x_T m_T,$$

and the ion-ion collision frequency summed over D and T,

$$\nu_{ii} = \sum_{s \in \{D, T\}} 4.80 \times 10^{-8} \frac{n_s [\text{cm}^{-3}] \ln \Lambda}{\sqrt{\mu_s} T_i [\text{eV}]^{3/2}}, \quad \mu_s = m_s / m_p.$$

Since  $\tau_{E,i} \propto T_i^{1/2}$ , the ion balance becomes a 1-D fixed-point root-find for  $T_i$  rather than a closed-form substitution. Pure neoclassical transport has no anomalous channel, so it under-estimates real ion transport and can predict very hot, sometimes implausible  $T_i$ ; treat it as a lower bound. It cannot be combined with electron-ion equipartition.

#### 1.10.6 Neoclassical ion (arbitrary aspect ratio)

The bare  $\varepsilon^{-3/2}$  geometric factor above both diverges as  $\varepsilon \rightarrow 0$  and stays bounded as  $\varepsilon \rightarrow 1$ , neither of which is right on a spherical tokamak. The *arbitrary A* variant (selector entries ending “*neoclassical i (arbitrary A)*”) replaces the trapped-fraction part with the Lin-Liu & Miller  $f_t/f_c$  the banana-regime structure of Helander & Sigmar’s *Collisional Transport in Magnetized Plasmas*, Ch. 11 and uses the arbitrary- $A$  edge safety factor  $q_a$ :

$$\chi_{i,\text{neo}} \simeq q_a^2 \rho_i^2 \nu_{ii} \frac{f_t}{f_c}, \quad f_t = 1 - \frac{(1 - \varepsilon)^2}{(1 + 1.46 \sqrt{\varepsilon}) \sqrt{1 - \varepsilon^2}}, \quad f_c = 1 - f_t,$$

$$q_a = \frac{5 a^2 B_t}{R_0 I_p [\text{MA}]} \frac{1 + \kappa^2 (1 + 2\delta^2 - 1.2\delta^3)}{2} \frac{1.17 - 0.65 \varepsilon}{(1 - \varepsilon^2)^2}.$$

$f_t/f_c$  stays finite for every  $\varepsilon < 1$  and rises steeply only as  $\varepsilon \rightarrow 1$  (all ions trapped) the physically correct low-aspect trend. This is **not** a smooth reduction of the  $\varepsilon^{-3/2}$  form: the two order-of-magnitude estimates can differ by up to about one order of magnitude in either direction (more once the  $\tau_{E,i} \propto T_i^{1/2}$  fixed point amplifies it), which is exactly why both are offered run them side by side. References: Helander, *Phys. Plasmas* **7** (2000) 3999; Hinton, Wiley *et al.*, *Phys. Rev. Lett.* **29** (1972) 698; Satake *et al.*, *Phys. Plasmas* **9** (2002); Goldston & Rutherford (1995).

#### 1.11 Mean fast-ion energy

Each beam has its own steady-state slowing-down distribution:

$$f_j(E) \propto \frac{\sqrt{E}}{E^{3/2} + E_{c,j}^{3/2}}, \quad 0 < E \leq E_{b,j}$$

Its mean fast-ion energy is:

$$\langle E_{fast} \rangle_j = \frac{\int_0^{E_{b,j}} E f_j(E) dE}{\int_0^{E_{b,j}} f_j(E) dE}$$

### 1.12 Combined values shown in Results

The GUI reports power-weighted effective values for the two beams.

Effective slowing-down time:

$$\tau_S^{\text{eff}} = \frac{\sum_j P_{\text{useful},j} \tau_{S,j}}{\sum_j P_{\text{useful},j}}$$

Effective mean fast-ion energy:

$$\langle E_{fast} \rangle = \frac{\sum_j P_{\text{useful},j} \langle E_{fast} \rangle_j}{\sum_j P_{\text{useful},j}}$$

These effective values are used for the combined fast-ion diagnostics, including fast-ion energy density, pressure, beta, and:

$$R = \frac{u_{fast}}{U_t}$$

The underlying HotJass calculation still evaluates each beam separately before forming these combined diagnostics.

### 1.13 Results tabs (Operating point)

- **Dashboard** scalar table of the operating point (temperatures, and their on-axis values when profile-corrected 0-D is on; densities, power balance terms, orbit widths, D-T / D-D fusion power, neutron rate, Greenwald ratio, edge safety factor  $q_*$ , beta).  $q_*$  is reported using whichever formula the active orbit-loss model actually uses – the bare cylindrical `safety_factor_cyl_edge` for "large-aspect", or the arbitrary- $A$  `safety_factor_cyl_edge_arbitrary_A` ( $q_a$ ) for either ST orbit variant – so it stays consistent with the orbit widths shown just above it. Per-beam CX-loss rows show  $f_{\text{cx},P}/f_{\text{cx},N}/\gamma_{\text{cx}}$  (plus  $n_0$  and the informational escape probability) when a physics-based CX model is selected, or just the flat fraction for "Manual fraction" – see "Charge-exchange loss" above. The rotation row shows "off" unless a Rotation model is selected, in which case it shows  $v_\phi$ , a Mach number, and (momentum-balance mode) the net NBI torque – see "Bulk toroidal rotation" above.
- **Deposition** (1) beam targeting geometry in the torus top view (titled with the active device), tangent to each beam's  $R_t$  with its co-/counter-current sense; (2) neutral-beam survival  $I(s)/I_0$  and the fast-ion birth rate  $n_e \sigma I(s)/I_0$  along the beam, annotated with each beam's shine-through percentage; (3) the resulting fast-ion birth versus normalised radius  $\rho$ , per beam and total (chord-sampled, midplane circular approximation), with the prompt first-orbit-loss zone  $\rho > \rho_{\text{cut}}$  shaded and a per-beam  $\rho_{\text{cut}}$  line (from the same criteria

as the orbit-loss model in use); (4) the steady-state slowing-down distribution  $f(E)$  with the  $E_b$  edge marked.

- **Power flow** waterfall / Sankey / pie of injected power to its sinks.
- **Profiles**  $n_e(\rho)$ ,  $T_{e,i}(\rho)$ ,  $\tau_S(\rho)$  (local thermalization time, from the same local  $n_e(\rho)$ ,  $T_e(\rho)$  used elsewhere in this panel – log scale, since it rises sharply toward the edge as  $n_e \rightarrow 0$ ),  $n_0(\rho)$  (background-neutral density behind the active CX-loss model – see "Charge-exchange loss" above), the plasma shape, and  $P_{fus}(\rho)$  (local D-T + D-D fusion power density, thermal vs. beam-plasma / beam-target, built from the same on-axis densities and  $(1 - \rho^2)^{2p}$  shapes as the volume-integrated totals on the Dashboard – see "Fusion power" below).
- **Fusion** a pie chart of  $P_{fus}$  split into its six channel/reaction combinations (D-T and D-D, each thermal / beam-target / beam-beam), with MW values and percentages; slices under 0.5% of the total are hidden to keep the chart readable (still fully available in the Dashboard's D-T/D-D breakdown rows). Title also shows  $Q$  and the full neutron rate  $Y_n$ .
- **Assumptions** the full validity / fit-range read-out and any warnings.
- **References** the literature behind the *currently selected* models (beam stopping, confinement scaling, orbit-loss model) and the active machine geometry, each with a link, plus a feedback button.

### 1.14 Electron-ion exchange time

The electron-ion exchange time is based on the NRL Plasma Formulary thermal-equilibration rate. For each thermal ion species  $s$  (D or T):

$$\bar{\nu}_{ei,s} = 1.8 \times 10^{-19} \frac{\sqrt{m_e m_s} Z_e^2 Z_s^2 n_e \ln \Lambda}{(m_s T_e + m_e T_i)^{3/2}}$$

In this practical-unit expression, masses are in grams, temperatures are in eV, and electron density is in  $\text{cm}^{-3}$ . The model uses  $Z_e = Z_s = 1$  for hydrogenic D and T ions.

For a D-T plasma, the effective exchange time is:

$$\tau_{IE} = \frac{n_D + n_T}{n_D \bar{\nu}_{ei,D} + n_T \bar{\nu}_{ei,T}}$$

For a single thermal ion species, this reduces to:

$$\tau_{IE} = \frac{1}{\bar{\nu}_{ei}}$$

The equivalent energy-over-power form used by the application is:

$$\tau_{IE} = \frac{\frac{3}{2}(n_D + n_T)|T_i - T_e|(10^3 e)V}{|P_{ie}|}$$

where  $V$  is the plasma volume and  $e = 1.602176634 \times 10^{-19}$  J/eV. The exchange power is:

$$P_{ie} = -P_{ei} = -\frac{3}{2}V(T_e - T_i)(n_D \bar{\nu}_{ei,D} + n_T \bar{\nu}_{ei,T})(10^3 e)$$

The absolute values make  $\tau_{IE}$  positive regardless of whether electrons heat ions ( $T_e > T_i$ ) or ions heat electrons ( $T_i > T_e$ ).

## 1.15 Fusion power

The total fusion power is the sum of the thermal D-T and beam-target contributions:

$$P_f = P_{f,\text{thermal}} + P_{f,\text{beam}}$$

### 1.15.1 Thermal D-T fusion

The thermal contribution is calculated from the thermal deuterium and tritium densities and the ion-temperature-dependent Bosch-Hale reactivity:

$$P_{f,\text{thermal}} = n_{D0} n_{T0} \langle \sigma v \rangle_{DT}(T_i) V E_f$$

Here  $n_{D0}$  and  $n_{T0}$  are the thermal species densities,  $V$  is the plasma volume, and:

$$E_f = 17.6 \text{ MeV} = 17.6 \times 10^6 e$$

The Bosch-Hale thermal reactivity is evaluated at  $T_i$ , because the relative velocity is that of the thermal ions, not the electrons.

### 1.15.2 Beam-target fusion

Each useful beam is evaluated separately against the stationary thermal ions of the other D-T species:

$$P_{f,\text{beam}} = \sum_j V E_f \int_0^{E_{b,j}} n_{\text{target},j} f_j(E) [\sigma v]_j(E) dE$$

The beam slowing-down distribution is normalized to the fast-ion density of that beam:

$$f_j(E) \propto \frac{\sqrt{E}}{E^{3/2} + E_{c,j}^{3/2}}, \quad \int_0^{E_{b,j}} f_j(E) dE = n_{b0,j}$$

The beam-target reactivity is:

$$[\sigma v]_j(E) = \sigma_{DT}(E) v_j(E), \quad v_j(E) = \sqrt{\frac{2E}{m_j}}$$

For a D-T beam-target reaction  $E_f = 17.6 \text{ MeV}$ . For a D beam,  $n_{\text{target},j} = n_{T0}$ ; for a T beam,  $n_{\text{target},j} = n_{D0}$ . The beam fast-ion density is obtained from the useful beam power:

$$n_{b0,j} = \frac{P_{\text{useful},j} T_{S,j}}{E_{b,j} (10^3 e) V}$$

The input shine-through fraction is used to determine captured beam power; the plotted  $P_{\text{shine}}$  is then the derived shine-through power from the scan. Charge-exchange and other configured losses similarly reduce the useful power before calculating  $n_{b0,j}$ .

The fusion post-processing always uses the  $(1 - \rho^2)$  profile weighting: the thermal contribution is

$$P_{f,\text{thermal}} = V E_f \langle n_D(\rho) n_T(\rho) \langle \sigma v \rangle_{DT} [T_i(\rho)] \rangle_V,$$

and the beam-target contribution uses the local target density and the local slowing-down distribution evaluated with  $T_e(\rho)$ . The  $n_{D0}$ ,  $n_{T0}$ ,  $T_e$ ,  $T_i$  used here are the **on-axis** values (see *Profile-corrected 0-D*).

### 1.15.3 D-D fusion

The D-D channel (present whenever the deuterium fraction  $x_D > 0$ ) is evaluated with the Bosch-Hale D-D fits, **branch-resolved**:

| branch                            | products                                         | $E_f$     | neutron  |
|-----------------------------------|--------------------------------------------------|-----------|----------|
| D(d,p)T                           | T (1.01) + p (3.02 MeV)                          | 4.033 MeV | none     |
| D(d,n) $\text{\text{3}}\text{He}$ | $\text{\text{3}}\text{He}$ (0.82) + n (2.45 MeV) | 3.269 MeV | 2.45 MeV |

**Thermal:**

$$P_{f,\text{DD}}^{\text{th}} = V [E_{ddp} \langle \frac{1}{2} n_D^2 \langle \sigma v \rangle_{ddp}(T_i) \rangle_V + E_{ddn} \langle \frac{1}{2} n_D^2 \langle \sigma v \rangle_{ddn}(T_i) \rangle_V],$$

with the  $\frac{1}{2}$  identical-particle factor and the same  $(1 - \rho^2)$  profile weighting as the D-T thermal term.

**Beam-target** (a fast D ion on the thermal-D population **no**  $\frac{1}{2}$  factor, the two populations are distinct):

$$P_{f,\text{DD}}^{\text{beam}} = \sum_{j \in \text{D beams}} V \int_0^{E_{b,j}} n_{D0} f_j(E) [E_{ddp} \sigma_{ddp}(E) + E_{ddn} \sigma_{ddn}(E)] v_j(E) dE.$$

The cross-sections take the **lab-frame** deuteron energy (same convention as  $\sigma_{DT}$ ). D-D does **not** produce alphas, so the alpha self-heating fixed-point closes on the D-T power only ( $P_\alpha = f_\alpha(3.5/17.6)P_{f,DT}$ ).

### 1.15.4 Neutron rate

$$Y_n = \frac{P_{f,DT}}{17.6 \text{ MeV}} + R_{ddn}^{\text{th}} + R_{ddn}^{\text{beam}} (+ Y_n^{\text{bb}}),$$

i.e. one 14.06 MeV neutron per D-T reaction plus the 2.45 MeV neutrons from the D(d,n) branch, plus the beam-beam contribution when that's enabled (see "Beam-beam fusion" above). Named  $Y_n$ , not  $R_n$ , to avoid confusion with the fast/thermal energy-density ratio  $R$  reported elsewhere. Reported on the Dashboard, broken down by channel (beam-target / thermal / beam-beam / full) in the Scan tab's `Y_beam`, `Y_thermal`, `Y_bb`, `Y_full` group and the Summary tab's "Neutron rate" panel, and as a slice of the Fusion tab's pie chart.

### 1.15.5 Current model limitations

- The beam-target integral uses stationary target ions, so the solved  $T_i$  is not included in that reaction integral, and the beam is treated as a single full-energy component (no half/third fractions).
- The tritium and  $\text{\text{3}}\text{He}$  bred by D-D are not burned further ("cat-DD" is neglected).
- The radial volume weighting uses the model's elliptical flux-surface approximation and does not yet solve a 2D equilibrium.

## 1.16 Density scan

The Plasma tab defines the requested scan range with `n_e_min` and `n_e_max`. The Results tab reports the physically valid interval separately. Points below the HotJass charge-neutrality feasibility limit are not assigned physical output values and are omitted from plots.

The  $T_e, T_i$  scan plot's y-axis top is  $\min(\max(T_e, T_i), 100 \text{ keV})$  – the actual data maximum, unless it exceeds 100 keV (the upper edge of the Bosch-Hale fits' validated range), in which case the axis is capped there so one runaway hot-ion point doesn't wash out the rest of the curve. The bottom is fixed at 0 (not autoscaled) so a fixed top never stretches the axis into an apparent negative margin.

The  $n_D, n_T, n_{b0}$  scan group's otherwise-empty 4th panel shows the fast-ion / target-ion density ratio  $n_{b0}/n_{\text{Target}}$ , where  $n_{\text{Target}}$  is whichever thermal D-T species the *dominant* (by injected power) NBI beam reacts with in the beam-target channel –  $n_T$  for a D beam,  $n_D$  for a T beam.

Four further scan groups:

- $P_e, P_i, P_{ie}, Q$  ("Powers") –  $Q = P_{f,tot}/P_{NB}$  replaces the flat shine-through power here (moved to "Losses" below, where it belongs with the other loss channels).
- $P_{f,tot}, P_{f,th}, P_{f,b}, P_{useful}$  ("Pf") – the previously-empty 4th panel now shows  $P_{useful} = P_{NB} - P_{lost}$ , the power that actually reaches the electron/ion balance.
- $P_{shine}, P_{orbit}, P_{cx}, P_{lost}$  ("Losses") – the three loss channels (whichever CX-loss model is active feeds  $P_{cx}$ ) plus their sum  $P_{lost} = P_{NB} - P_{useful}$ .
- $Y_{beam}, Y_{thermal}, Y_{bb}, Y_{full}$  ("Neutrons") – the neutron rate split by reaction channel: beam-target (D-T + D-D), thermal-thermal (D-T + D-D), beam-beam (Sec. "Beam-beam fusion" above – a flat zero while that's off), and the full total ( $Y_{beam} + Y_{thermal} + Y_{bb} = Y_{full}$  exactly). Named  $Y$ , not  $R$ , so it isn't confused with the  $R = u_{fast}/U_{th}$  scan group.

## 1.17 Important notation

- $T_e$ : electron temperature
- $T_i$ : thermal-ion temperature
- $P_e$ : beam power deposited into electrons
- $P_i$ : beam power deposited into ions
- $L_e, L_i$ : beam-power electron / ion heating fractions ( $L_e + L_i = 1$ )
- $P_{ie}$ : electron-ion energy exchange power
- $P_{ECRH}, P_{ICRH}$ : ECRH and ICRH auxiliary heating power
- $P_{aux,e}, P_{aux,i}$ : auxiliary heating to electrons / ions
- $P_\alpha, P_{\alpha,e}, P_{\alpha,i}$ : fusion alpha self-heating power (total, to electrons, to ions)
- $\tau_{E,e}, \tau_{E,i}$ : electron / ion energy confinement times used in the balance
- $n_{thermal}$ : thermal-ion density,  $n_{sum}(n_{e0}, Z_{eff}) - n_{b0}$
- $n_D, n_T$ : thermal deuterium and tritium densities
- $n_{b0}$ : total fast-ion density
- $\rho_{Li}$ : fast-ion toroidal-field Larmor radius at the injection energy
- $\Delta r$ : passing drift-orbit radial width,  $q_* \rho_{Li}$  (large-aspect model)
- $\rho_\theta$ : fast-ion poloidal gyroradius,  $(B_t/\bar{B}_p) \rho_{Li}$  (ST orbit models)
- $w_{pass}, w_{ban}$ : ST circulating / banana orbit widths
- $f_t, f_c$ : trapped / circulating particle fractions (arbitrary  $A$ )
- $q_a$ : arbitrary-aspect-ratio edge safety factor (Uckan form)
- $f_{orbit}$ : first-orbit loss fraction of captured beam power
- $P_{orbit}$ : first-orbit loss power
- $P_{f,tot}, P_{f,th}, P_{f,b}$ : total, thermal, and beam-target fusion power (D-T)
- $P_{f,DT}, P_{f,DD}$ : D-T and D-D fusion power ( $P_{f,tot} = P_{f,DT} + P_{f,DD}$ )

- $R_n$ : neutron rate (14 MeV D-T + 2.45 MeV D-D branch)
- $\langle T \rangle, T_0$ : volume-averaged and on-axis temperature (profile-corrected 0-D)
- $p_n, p_{Te}, p_{Ti}$ : density and electron / ion temperature profile exponents  $((1 - \rho^2)^{2p})$
- $\tau_{IE}$ : effective electron-ion exchange time
- $p_{th}, p_{fast}$ : thermal and fast-ion pressure
- $\beta_t$ : toroidal beta